Faiza Ghous

Nationality: Pakistani **Date of birth:** 10/10/1997

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Home: Islamabad (Pakistan)

WORK EXPERIENCE

National Disaster Management Authority – Islamabad, Pakistan

Assistant Manager (Anticipatory Actions) - Disaster Risk Reduction

[01/03/2026 – Current]

- Design and implement disaster risk reduction (DRR) strategies to enhance community resilience and minimize hazard impacts.
- Support anticipatory action frameworks by integrating early warning systems and forecast-based decision-making into program planning.
- Conduct risk and vulnerability assessments to inform evidence-based policies and targeted interventions.
- Collaborate with stakeholders to develop and advocate for policy solutions aligned with climate adaptation and disaster preparedness goals.
- Strengthen institutional capacity through training, coordination, and knowledge sharing on resilience, preparedness, and proactive response mechanisms.

National Disaster Management Authority – Islamabad, Pakistan

Assistant Manager (Meteorology)

[01/11/2023 – 28/02/2026]

- Apply multi-model ensemble outputs and regional downscaling for disaster risk management
- Integrated meteorological projections into multi-hazard early warning systems, linking climate science to practical disaster preparedness.
- Generate short-term, mid-term, and long-term weather forecasts using meteorological models and observational data.
- Monitor severe weather events such as storms, heatwaves, and heavy rainfall, ensuring timely alerts.
- Analyse climate data to identify trends, anomalies, and potential long-term impacts.
- Coordinate with national disaster management teams to assess weather-related hazards and support early warning systems.

Weatherwalay – Islamabad, Pakistan

Weather Specialist

[01/04/2021 – 31/10/2023]

- Utilize numerical models, satellite data, and radars to provide accurate weather predictions.
- Provide location-specific weather predictions for critical projects and events.
- Applied ensemble modeling for uncertainty quantification in forecasts, linking probabilistic projections with adaptation planning.
- Collaborate with media to share clear and actionable weather information.

EDUCATION AND TRAINING

Climate Modelling-ICON Numerical Model Course 2024

Deutscher Wetterdienst (DWD), Headquarters [10/06/2024 – 14/06/2024]

City: Offenbach | **Country:** Germany

• **Comprehensive Training on Numerical Weather Prediction Models:** Focus on the structure, configuration, and operational applications of models like COSMO, ICON, and CLM.

• **Hands-on Sessions for Model Setup and Data Assimilation:** Learn model initialization, boundary conditions, and assimilation of real-time observational data.
• **Climate and Forecasting Applications:** Explore the use of numerical models for weather forecasting, climate simulations, and extreme weather event prediction.

M.Phil. Earth Sciences

Quaid-e-Azam University Islamabad [15/02/2020 – 31/05/2022]

City: Islamabad | **Country:** Pakistan | **Field(s) of study:** Natural sciences, mathematics and statistics: ● Earth sciences | **Final grade:** CGPA: 3.6 out of 4.0 | **Thesis:** Static Earthquake Triggering Of Central Himalaya in Pakistan Region By Coulomb Failure

- Analyzed static stress transfer mechanisms to evaluate earthquake triggering in the Central Himalayan region of Pakistan.
- Applied **Coulomb Failure Stress (CFS)** modeling to identify seismically vulnerable zones and potential aftershock regions.
- Assessed the implications of stress redistribution for seismic hazard evaluation and regional earthquake risk assessment.

Bachelor Earth Sciences

Quaid-e-Azam University Islamabad [26/10/2015 – 25/12/2019]

City: Islamabad | **Country:** Pakistan | **Field(s) of study:** Natural sciences, mathematics and statistics: ● Earth sciences | **Final grade:** CGPA: 3.3 out of 4.0 | **Thesis:** Seismic Interpretation of Lower Indus Basin, Bitrism Area, Pakistan

- Conducted seismic interpretation of 2D reflection data to delineate subsurface structural and stratigraphic features in the Lower Indus Basin.
- Identified key hydrocarbon bearing formations and fault systems through seismic attribute analysis and horizon mapping.
- Evaluated basin tectonics and depositional settings to support petroleum system assessment and exploration potential.

CONFERENCES AND SEMINARS

National & International Training Workshops on Disaster Risk Reduction

Organizer & Participant – Community Vulnerability & Hazard Preparedness Programs

- Organized and actively participated in multiple training sessions focused on understanding vulnerable communities and hazard preparedness.
- Facilitated workshops addressing risk communication, community resilience, and adaptive strategies.
- Promoted inclusive approaches to disaster management, emphasizing early action and preparedness.
- Coordinated with local and international stakeholders to enhance capacity building initiatives.

[04/03/2024 – 05/03/2024] International Conference on Climate Resilience and Early Warning Systems

Early Warning Systems & Disaster Preparedness

- Contributed as a subject expert on early warning systems for hazard mitigation and disaster risk reduction, building on technical knowledge gained through Japan International Cooperation Agency training programs.
- Delivered insights on integrating geoscience data into forecasting and community alert mechanisms.
- Engaged with international researchers and policymakers to strengthen multi-hazard early warning frameworks.
- Supported knowledge exchange on improving preparedness strategies in vulnerable regions.

[06/04/2025 – 07/04/2025] Japan International Cooperation Agency & US Army Corps of Engineers Collaborative Training Program

Capacity Building & Hydrological Modelling Training

- Completed specialized training in hydrological modelling using HEC-HMS for flood forecasting and watershed analysis.
- Strengthened technical skills in rainfall-runoff modelling and disaster risk assessment.
- Participated in capacity-building sessions focused on sustainable water resource management.
- Collaborated with international experts to apply modelling techniques in real-world hazard scenarios.

[01/02/2023 – 06/2023]

AAPGWN-2023 Member (American Association of Petroleum Geologist Women Network)

- **Global Networking Opportunities:** Connect with professionals, researchers, and industry leaders in geosciences, with a focus on empowering women in the energy sector.
- **Leadership and Career Development Programs:** Participate in mentoring sessions, workshops, and webinars tailored to advancing women's roles in petroleum geology and related fields.
- **Promoting Diversity and Inclusion:** Contribute to initiatives that foster an inclusive environment for women in geosciences and address gender-based challenges in the industry.
- **Access to Industry Insights and Resources:** Stay updated with the latest research, trends, and innovations in petroleum geology through exclusive publications and events.

PROJECTS

CCOP Dashboard- Centoriun Climate Operating Picture

- Utilizes CMIP6 global climate models to simulate future climate projections from 2000 to 2100 based on SSPs (Shared Socioeconomic Pathways).
- Integrates temperature and precipitation variables to assess regional and global-scale climatic trends and anomalies.
- Detects and visualizes the frequency, intensity, and spatial distribution of extreme weather events including heatwaves, droughts, and heavy precipitation.
- Enables scenario-based climate impact assessments using multiple SSP trajectories such as SSP1-2.6, SSP2-4.5, and SSP5-8.5.
- Supports climate resilience and adaptation planning by offering high-resolution outputs for both near-term (2021–2040) and long-term (2081–2100) horizons.

Probabilistic Meteorological Projection by Multi Model Ensemble CMIP6 (Regional and Local)

- Probabilistic Meteorological Projection by Multi-Model Ensemble CMIP6 involves using outputs from multiple climate models to create a range of future climate scenarios.
- This approach reduces uncertainties compared to single-model predictions.
- Probabilistic projections provide a spectrum of possible outcomes with associated probabilities.
- These projections aid in more informed decision-making.
- Down-scaling these projections to regional and local levels makes the analysis more relevant for specific areas.
- This enhanced analysis supports better planning and adaptation strategies for climate impacts.

Impact-based Forecasting (Islamabad)

- Impact-based forecasting in Islamabad focuses on predicting the potential consequences of weather events to improve preparedness and response.
- This approach integrates hazard predictions with vulnerability and exposure data to assess risks and communicate actionable information to decision-makers and the public.
- By tailoring forecasts to highlight specific impacts, such as flooding or heatwaves, impact-based forecasting enhances the effectiveness of early warning systems and risk management strategies.

Seismic Events in Active Tectonic Areas Under Climate Change

- Conduct interdisciplinary research and develop predictive models on the impacts of tectonic activity and climate change.
- Analyse seismic and climate data to identify inter-dependencies and provide disaster preparedness recommendations.
- Integrate and analyse seismic and climatic data to create predictive models and develop data visualisation Tools.
- Apply numerical models to predict seismic and climatic events.

RESEARCH EXPOSURES

Helmholtz-Zentrum Hereon, Geesthacht Germany

- **Exposure to Climate and Earth System Research:** Engaged with advanced research in regional climate modeling, sea-level rise, and ocean-atmosphere interactions.
- **Hands-on Experience with High-Performance Computing (HPC):** Utilized cutting-edge computational tools and numerical models for environmental data analysis and simulations.
- **Collaboration on Sustainability and Adaptation Projects:** Contributed to interdisciplinary projects addressing climate adaptation, coastal management, and environmental risk assessments.

Potsdam Institute for Climate Impact Research

- **Climate Data Modeling and Analysis:** Utilized high-resolution Earth system models to simulate climate impacts and assess regional vulnerabilities.
- **Big Data Processing and Visualization:** Applied advanced techniques for handling large datasets using Python, MATLAB, and GIS tools to analyze climate scenarios and trends.
- **Model Development and Calibration:** Gained expertise in calibrating integrated assessment models (IAMs) and Earth system models for precise climate forecasting and impact studies.

GERICS Climate Service Center Germany

- **Development of Climate Services and Tools:** Designed user-oriented climate products and decision-support tools tailored for adaptation planning and policy development.
- **Regional Climate Modeling and Projections:** Worked with downscaling techniques to produce localized climate projections using state-of-the-art multi-model ensembles.
- **Climate Risk Assessment and Impact Studies:** Conducted technical assessments of sectoral risks (e.g., agriculture, water resources) under various climate scenarios to support sustainable solutions.

GFZ German Research Centre for Geosciences

- **Seismic Hazard and Crustal Stress Analysis:** Worked with geophysical models to study fault mechanics, stress accumulation, and earthquake triggering.
- **Satellite and Geodetic Data Applications:** Analyzed InSAR and GNSS data for monitoring tectonic movements, land deformation, and early warning systems.
- **High-Performance Computing for Geophysical Simulations:** Utilized advanced computational tools to simulate geodynamic processes and assess natural hazards.

DWD Deutscher Wetterdienst, Offenbach am Main, Germany

- **Numerical Weather Prediction (NWP) and Climate Modeling:** Gained expertise in using models like ICON and COSMO for weather forecasting and climate projections.
- **Data Assimilation and Forecasting Systems:** Worked with real-time observational data integration to improve weather forecasts and early warning systems.
- **Extreme Weather Event Analysis:** Conducted assessments of high-impact weather phenomena to enhance disaster preparedness and mitigation strategies.

Federal Institute for Geosciences and Natural Resources (BGR)

- **Geoscientific Research and Resource Management:** Engaged in projects focused on the sustainable use of natural resources and environmental monitoring.
- **Geophysical Surveys and Data Analysis:** Conducted geophysical investigations, including seismic and magnetic surveys, to assess subsurface structures and resource potential.
- **Collaboration on Geological Hazard Assessments:** Contributed to interdisciplinary studies evaluating geological hazards, including landslides and subsidence, for risk mitigation and management.

TECHNICAL SKILLS

Multi Model Ensemble

Projections by incorporating **CMIP5/6** Multi Models and various **SSP** for better and weather forecast.

Climate Data Operator (CDO)

Command in manipulating and processing climate datasets (e.g., netCDF, GRIB), including data regridding, interpolation, statistical analysis, and time-series extraction for advanced climate modeling and projections.

Satellite Imagery Analysis

Skilled in analyzing satellite imagery to identify and interpret atmospheric patterns, cloud formations, and other meteorological phenomena.

Radar Data Interpretation

Experience in interpreting radar data to track and analyze precipitation, severe weather, and other atmospheric conditions affecting local and regional areas.

Impact based Weather Forecasting

Risk Assessments and planning of vulnerable areas affected by weather induced disasters.

Meteorological Projections

Seasonal, Monthly, Weekly, and Daily Projections and Advisories.

Communication of Forecasts

Effective communication of weather forecasts to various stakeholders, including
internal teams, emergency respondents, and the public through written reports, presentations, and verbal communication.

RECOMMENDATIONS

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